

MP1994V2 Proof Ledger

Mortensen–Pissarides (1994) Lean replication

4 August 2026

Purpose. The milestone tracker records project state. This cumulative ledger records exactly what Lean has proved, the hypotheses used, and the economic meaning of each theorem.

1 M1: Surplus Bellman equation

Paper reference Section 3, journal page 401, equation (8)
Adequacy status **COMPLETE – GREEN**
Lean source MP1994V2/SteadyState/Surplus.lean

1.1 Mathematical statement

For the actual equilibrium surplus

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U,$$

Equation (8) is a conditional representation theorem: for every object satisfying `ValueEquilibrium P`, Lean proves that the surplus derived from its value functions satisfies, for every idiosyncratic state ε ,

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \lambda \int \max\{S(x), 0\} dF(x) - \beta\theta q(\theta)S(\varepsilon_u). \quad (8)$$

Here `P.shock` is the probability law of the idiosyncratic shock; its induced CDF is F , and integration with respect to dF is represented in Lean by integration against `P.shock`. Also, $\theta q(\theta)$ is `P.workerMeetingRate E.theta`. M1 does not prove existence of a `ValueEquilibrium`.

1.2 Economic interpretation

- $p + \sigma\varepsilon$: current match output.
- b : unemployment opportunity cost.
- $\lambda \int \max\{S(x), 0\} dF(x)$: option value of future idiosyncratic draws, allowing rejection of nonpositive continuation surplus.
- $\beta\theta q(\theta)S(\varepsilon_u)$: the employed worker’s foregone search gain at the upper-support job.

1.3 Exact Lean declarations

Role	Declaration
Continuation decomposition	MP1994V2.continuationTerm_eq_integral_activeSurplus_sub
Firm surplus share	MP1994V2.ValueEquilibrium.firm_share
Algebraic surplus flow	MP1994V2.ValueEquilibrium.surplus_flow_equation
Minimal probability theorem	MP1994V2.ValueEquilibrium.surplus_bellman_of_probability
Paper-level wrapper	MP1994V2.ValueEquilibrium.surplus_bellman

1.4 Assumptions

Used:

- equation (3), through the definition of actual surplus;
- equation (4), only at the upper state;
- equations (5) and (6), added to cancel the wage;
- equation (7), subtracted to account for unemployment value;
- probability normalization of `P.shock`;
- `ValueEquilibrium.active_surplus_integrable`.

The minimal theorem assumes only `[IsProbabilityMeasure P.shock]`. The wrapper takes `D : ShockAssumptions P`, installs `D.isProbability` locally, and uses no other field of `D`.

Not used: parameter signs; `CoreEconomicAssumptions`; upper-support restrictions or maximality; atomlessness; mean zero; unit variance; `ShockNormalizationAssumptions`; matching positivity or monotonicity; `MatchingAssumptions`; equations (1) and (2).

1.5 Readable proof

Write the shared continuation term as

$$C(\varepsilon) = \int [\max\{S(x), 0\} - S(\varepsilon)] dF(x).$$

Adding equations (5) and (6) cancels the wage and sums the Nash weights:

$$r[J(\varepsilon) + W(\varepsilon)] = p + \sigma\varepsilon + \lambda C(\varepsilon).$$

Subtracting equation (7), and using equation (3), gives

$$rS(\varepsilon) = p + \sigma\varepsilon - b + \lambda C(\varepsilon) - \theta q(\theta)[W(\varepsilon_u) - U].$$

Equation (4) at ε_u gives

$$W(\varepsilon_u) - U = \beta S(\varepsilon_u).$$

Probability normalization of `P.shock` and active-surplus integrability give

$$C(\varepsilon) = \int \max\{S(x), 0\} dF(x) - S(\varepsilon).$$

Substitution and moving $-\lambda S(\varepsilon)$ to the left proves (8).

1.6 Adequacy and next dependency

The actual `ValueEquilibrium` surplus is used. Equation (8) is derived, not added as an assumption or field. No cutoff, reservation rule, or affine surplus result is used. It is conditional on an existing `ValueEquilibrium` and does not establish that such an equilibrium exists.

Human-review conclusion. Equation (8) matches the paper and is derived non-circularly from equations (3)–(7). The actual equilibrium surplus is used, with no cutoff, affinity result, or reduced equilibrium. The theorem is explicitly conditional on an existing `ValueEquilibrium`.

M2 uses (8) as the reviewed dependency for the affine-surplus and reservation results recorded next.

2 M2: Affine surplus and endogenous reservation cutoff

Paper reference	Section 3; journal p. 401, reservation-property paragraph; journal p. 402, equation (12)
Adequacy status	COMPLETE – GREEN
Lean sources	MP1994V2/SteadyState/AffineSurplus.lean; MP1994V2/SteadyState/Cutoff.lean

2.1 Mathematical statements

For every existing `ValueEquilibrium P`, subtracting equation (8) at two states gives

$$(r + \lambda) [S(\varepsilon_2) - S(\varepsilon_1)] = \sigma(\varepsilon_2 - \varepsilon_1).$$

Under $r > 0$, $\lambda \geq 0$, and $\sigma > 0$,

$$S(\varepsilon_2) - S(\varepsilon_1) = \frac{\sigma}{r + \lambda}(\varepsilon_2 - \varepsilon_1),$$

so S is strictly increasing. Define

$$\varepsilon_d = \varepsilon_u - \frac{(r + \lambda)S(\varepsilon_u)}{\sigma}.$$

Lean proves a unique zero on all of $\mathbb{R}$:

$$S(\varepsilon_d) = 0, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d.$$

The reservation rules are

$$\begin{aligned} S(\varepsilon) < 0 &\iff \varepsilon < \varepsilon_d, & 0 \leq S(\varepsilon) &\iff \varepsilon_d \leq \varepsilon, \\ J(\varepsilon) < 0 &\iff \varepsilon < \varepsilon_d, & 0 \leq J(\varepsilon) &\iff \varepsilon_d \leq \varepsilon. \end{aligned}$$

Paper equation (12) and its zero-anchored form are

$$S(\varepsilon) - S(\varepsilon_d) = \frac{\sigma(\varepsilon - \varepsilon_d)}{r + \lambda}, \quad S(\varepsilon) = \frac{\sigma}{r + \lambda}(\varepsilon - \varepsilon_d).$$

The M3 bridge is

$$\max\{S(\varepsilon), 0\} = \frac{\sigma}{r + \lambda} \max\{\varepsilon - \varepsilon_d, 0\}.$$

2.2 Proof strategy and economic separation

The expectation and upper-job search term in equation (8) are common across current states, so they cancel when the two equation instances are subtracted. The resulting positive affine slope constructs the unique mathematical zero.

Economic admissibility is proved separately. Equation (2) and $r > 0$ give $V = 0$; equation (1) then gives

$$q(\theta)J(\varepsilon_u) = c.$$

Positive vacancy cost and meeting imply $J(\varepsilon_u) > 0$. Since $1 - \beta > 0$, the upper-state surplus is positive, and therefore $\varepsilon_d < \varepsilon_u$. Recruiting cost thus places the reservation state strictly below the productivity of a newly formed upper-support job.

2.3 Exact Lean declarations

Role	Declaration
Scaled two-point identity	ValueEquilibrium.surplus_difference_scaled_of_probability
Divided affine identity	ValueEquilibrium.surplus_difference
Strict monotonicity	ValueEquilibrium.surplus_strictMono
Derived cutoff	ValueEquilibrium.reservationCutoff
Cutoff zero	ValueEquilibrium.surplus_reservationCutoff_eq_zero
Unique zero	ValueEquilibrium.existsUnique_surplus_zero
Vacancy value	ValueEquilibrium.vacancy_value_eq_zero
Free-entry product	ValueEquilibrium.vacancy_meeting_mul_upper_firm_value_eq_cost
Upper firm value	ValueEquilibrium.upper_firm_value_pos
Upper surplus	ValueEquilibrium.upper_surplus_pos
Cutoff below upper point	ValueEquilibrium.reservationCutoff_lt_epsUpper
Equation (12)	ValueEquilibrium.equation12
Surplus sign rule	ValueEquilibrium.surplus_neg_iff_lt_reservationCutoff
Firm sign rule	ValueEquilibrium.firm_value_neg_iff_lt_reservationCutoff
Active-surplus bridge	ValueEquilibrium.activeSurplus_eq_slope_mul_positivePart
Capstone	ValueEquilibrium.milestone2_capstone

Capstone zero identification. The capstone explicitly includes

$$\forall \varepsilon, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d,$$

rather than merely asserting an unidentified unique zero.

2.4 Assumptions

Core bundle. The affinity and cutoff theorems formally take $A : \text{CoreEconomicAssumptions } P$. Within that bundle their proof terms access `r_pos`, `lambda_nonneg`, and `sigma_pos`. Economic admissibility additionally accesses `c_pos` and `beta_lt_one`, as well as equations (1)–(2) and equilibrium $\theta > 0$.

Matching bundle. Paper-facing admissibility and capstone theorems formally take

$$M : \text{MatchingAssumptions } P.$$

Their proof terms access only `vacancyMeetingRate_pos`. The monotonicity fields remain part of the bundled theorem premise even though they are not referenced by the proof term.

Shock bundle. Wrappers taking $D : \text{ShockAssumptions } P$ access only `D.isProbability` in the `M2` affinity and zero arguments. The remaining distributional fields are carried by the paper-level bundle but are not accessed by those proof terms.

Thus the formal paper-facing premises include the complete bundles where shown in the signatures. Within them, the proofs do not access atomlessness, upper-support maximality, shock moment normalization, matching monotonicity, or any elasticity or differentiability restriction.

2.5 Equation (12) dependency

`equation12` follows from the affine-difference theorem evaluated at the derived cutoff. The separate zero theorem is required to simplify (12) to

$$S(\varepsilon) = \frac{\sigma}{r + \lambda}(\varepsilon - \varepsilon_d).$$

2.6 Interpretation, limitations, and next dependency

The positive slope means that higher idiosyncratic productivity raises joint match surplus by the constant factor $\sigma/(r + \lambda)$. The unique zero is the reservation productivity: firms destroy matches below it, are indifferent at it, and retain matches at or above it.

These results are conditional on an existing **ValueEquilibrium**. M2 does not prove existence of that equilibrium or of the joint pair (θ, ε_d) . It proves a unique zero on $\mathbb{R}$ and that the zero lies below ε_u , but not that it belongs to the topological support of the shock law. No lower-support assumption is present; the cutoff may lie below the lower effective support, making destruction probability zero.

Human-review conclusion. The surplus-affinity proof is non-circular, and the cutoff is derived from actual equilibrium values. Mathematical zero existence and economic admissibility remain separate; equations (1)–(2) are used only for admissibility. No M3 result is assumed or proved. The reviewed capstone explicitly identifies every surplus zero with **reservationCutoff**.

M3 may use the active-surplus positive-part formula to derive the continuation-value identity and the job-destruction and job-creation equations. No such integral identity is proved in M2.

3 M3: Continuation value, job destruction, and job creation

Paper reference	Section 3; journal page 401, equations (9)–(10); journal page 402, equation (13)
Adequacy status	COMPLETE – GREEN
Lean sources	SteadyState/Continuation.lean SteadyState/JobDestruction.lean SteadyState/JobCreation.lean
Capstone source	SteadyState/StaticConditions.lean

3.1 Definitions and continuation identity

The foundational expected excess and paper-facing tail option value are

$$H(d) = \int \max\{x - d, 0\} dP.\text{shock}(x), \quad T(d) = \int_d^{\varepsilon_u} [1 - F(x)] dx.$$

The CDF F is induced by **P.shock**; it is not a separate primitive. Lean also defines measure and tail forms of the job-destruction residual, the paper-form job-creation residual, and zero predicates for those residuals.

M2 supplies

$$\max\{S(x), 0\} = \frac{\sigma}{r + \lambda} \max\{x - \varepsilon_d, 0\}.$$

Integrating gives the foundational measure-form continuation identity

$$\int \max\{S(x), 0\} dP.\text{shock}(x) = \frac{\sigma}{r + \lambda} H(\varepsilon_d).$$

Integrability of the positive part is derived from **active_surplus_integrable** and the nonzero positive M2 slope; it is not added as a new assumption.

3.2 Strict-tail layer cake

Mathlib's `MeasureTheory.Integrable.integral_eq_integral_meas_lt` gives

$$H(\varepsilon_d) = \int_{t>0} P.\text{shock}\{x : t < \max(x - \varepsilon_d, 0)\} dt.$$

For $t > 0$, the level set is $(\varepsilon_d + t, \infty)$. Probability normalization proves

$$P.\text{shock}.\text{real}((y, \infty)) = 1 - F(y).$$

The almost-sure upper bound makes the tail vanish beyond ε_u . Truncation and the interval-translation lemma then give

$$H(\varepsilon_d) = T(\varepsilon_d).$$

The strict-tail theorem avoids the paper's informal differentiation and integration-by-parts wording. It also avoids atomlessness: no conversion between weak and strict tails is required.

3.3 Equations (9) and (10)

Substitution in equation (8) produces equation (9):

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \frac{\lambda\sigma}{r + \lambda}H(\varepsilon_d) - \beta\theta q(\theta)S(\varepsilon_u), \quad (9M)$$

and the paper form replaces $H(\varepsilon_d)$ by $T(\varepsilon_d)$.

Equations (1)–(2) and firm sharing give

$$\beta\theta q(\theta)S(\varepsilon_u) = \frac{\beta c}{1 - \beta}\theta.$$

Evaluating (9M) at the zero-surplus cutoff gives the measure form

$$p + \sigma\varepsilon_d = b + \frac{\beta c}{1 - \beta}\theta - \frac{\lambda\sigma}{r + \lambda}H(\varepsilon_d).$$

Replacing H by T is paper equation (10). The middle term is the worker's foregone search opportunity; the final subtraction is the option value of retaining a temporarily unprofitable job.

3.4 Equation (13)

Independently of equation (10), free entry, firm sharing, and M2 affine surplus at the new-job state give

$$q(\theta)(1 - \beta)\frac{\sigma}{r + \lambda}(\varepsilon_u - \varepsilon_d) = c.$$

Positivity of the denominator factors yields

$$q(\theta) = \frac{c}{1 - \beta} \frac{r + \lambda}{\sigma(\varepsilon_u - \varepsilon_d)}. \quad (13)$$

A smaller productivity gap raises the required vacancy meeting rate. Strict decrease of q implies lower market tightness, while the current weak monotonicity assumption on the worker meeting rate implies a weakly lower worker job-finding rate. M3 does not prove a comparative-static theorem across equilibria.

3.5 Exact Lean declarations

Role	Declaration
Expected excess	<code>Primitives.expectedExcess</code>
Tail probability	<code>Primitives.tailProbability</code>
Tail option value	<code>Primitives.tailOptionValue</code>
Strict-tail/CDF identity	<code>Primitives.measureReal_Ioi_eq_one_sub_cdf</code>
Excess integrability	<code>ValueEquilibrium.positivePart_sub_cutoff_integrable</code>
Layer-cake identity	<code>ValueEquilibrium.expectedExcess_cutoff_eq_tailOptionValue</code>
Active-surplus expectation	<code>ValueEquilibrium.integral_activeSurplus_eq_slope_mul_expectedExcess</code>
Equation (9), measure form	<code>ValueEquilibrium.equation9_measure</code>
Equation (9), paper form	<code>ValueEquilibrium.equation9</code>
Search gain	<code>ValueEquilibrium.search_gain_eq</code>
Equation (10), measure form	<code>ValueEquilibrium.equation10_measure</code>
Equation (10), paper form	<code>ValueEquilibrium.equation10</code>
Destruction residual, measure	<code>Primitives.jobDestructionResidualMeasure</code>
Destruction residual, tail	<code>Primitives.jobDestructionResidual</code>
Destruction witness, measure	<code>ValueEquilibrium.satisfiesJobDestructionMeasure</code>
Destruction witness, tail	<code>ValueEquilibrium.satisfiesJobDestruction</code>
Creation product	<code>ValueEquilibrium.job_creation_product_identity</code>
Equation (13)	<code>ValueEquilibrium.equation13</code>
Creation residual	<code>Primitives.jobCreationResidual</code>
Creation witness	<code>ValueEquilibrium.satisfiesJobCreation</code>
Capstone	<code>ValueEquilibrium.milestone3_capstone</code>

3.6 Assumptions and limitations

Paper-facing theorems formally carry the complete `CoreEconomicAssumptions`, `ShockAssumptions`, and `MatchingAssumptions` bundles.

- Measure-form continuation and equation (9) use probability normalization, active-surplus integrability, and the core fields `r_pos`, `lambda_nonneg`, and `sigma_pos`.
- The tail identity accesses `D.isProbability` and `D.upperSupport`. It does not access `D.noAtoms` or first-moment integrability.
- Its matching bundle is used through `reservationCutoff_lt_epsUpper`, whose proof accesses only `vacancyMeetingRate_pos`. Matching monotonicity remains in the formal bundle but is not referenced by the M3 proof term.
- Equation (10) additionally uses free entry and `beta_lt_one`. Its measure form requires no matching bundle.
- Equation (13) uses free entry, firm sharing, denominator positivity, and cutoff admissibility. Its shock bundle installs only probability normalization for the M2 affine formula.

Shock mean zero, unit variance, `ShockNormalizationAssumptions`, atomlessness, matching elasticity, differentiability, matching monotonicity, and comparative-static assumptions are deliberately unused.

M3 remains conditional on an existing `ValueEquilibrium`. It introduces no reduced-equilibrium structure, reverse reconstruction, joint-equilibrium existence or uniqueness, unemployment stock, or comparative-static theorem.

Human-review conclusion. The strict-tail layer-cake derivation is valid and does not use atomlessness. Equations (9), (10), and (13) match the paper, and equations (10) and (13) are derived independently. The residual results are forward conditional theorems for an existing `ValueEquilibrium`; no reduced equilibrium, reconstruction, or existence theorem is proved.

3.7 Residual admissibility warning

`SatisfiesJobDestruction` and `SatisfiesJobCreation` alone are not a complete reduced equilibrium. M4 must add $\theta > 0$, $d < \varepsilon_u$, and the relevant primitive, distributional, denominator, and other admissibility assumptions. The quotient-form job-creation residual must not be used without proving its denominator nonzero. The product-form job-creation identity is the robust foundational representation.

3.8 Dependency unlocked

M4 may define a `ReducedEquilibrium` only after supplying the missing admissibility contract. The current tail identity obtains integrability from an existing `ValueEquilibrium`. Reverse reconstruction requires a generic proof that $x \mapsto \max\{x - d, 0\}$ is integrable using the assumption

`ShockAssumptions.firstMomentIntegrable.`

M4 also needs, for an arbitrary admissible cutoff d , the generic identity

`expectedExcess d = tailOptionValue d,`

not only the specialization at `E.reservationCutoff`, before constructing value functions from a reduced pair.

4 M4: Reduced-equilibrium reconstruction and equivalence

Paper reference	Section 3, equations (10), (13), journal page 402
Adequacy status	COMPLETE – GREEN
Lean sources	<code>Equilibrium/Reduced.lean</code> ; <code>SteadyState/ReducedAnalytic.lean</code> ; <code>ForwardBridge.lean</code> ; <code>Reconstruction.lean</code> ; <code>Equivalence.lean</code>

4.1 Objects and robust conditions

`ValueEquilibrium P` is the primitive value system. `ReducedEquilibrium P` stores only $\theta > 0$, $d < \varepsilon_u$, measure-form job destruction, and product-form job creation. A full steady state additionally contains unemployment and is deferred to M6. Measure form integrates directly against `P.shock`; product form avoids division by $\varepsilon_u - d$. The CDF-tail equation (10) and quotient equation (13) are derived.

4.2 Generic analysis and reconstruction

`Primitives.positivePart_sub_integrable` uses probability normalization and first-moment integrability. The generic expected-excess/tail theorem additionally uses the almost-sure upper bound to identify the induced-CDF tail integral for every $d \leq \varepsilon_u$. Atomlessness is unused.

For a reduced pair R , Lean reconstructs

$$\begin{aligned}
 S_R(\varepsilon) &= \frac{\sigma}{r + \lambda}(\varepsilon - d), & V_R &= 0, & J_R &= (1 - \beta)S_R, \\
 U_R &= \frac{b + \beta\theta q(\theta)S_R(\varepsilon_u)}{r}, & W_R &= U_R + \beta S_R, \\
 w_R(\varepsilon) &= \beta[p + \sigma\varepsilon] + (1 - \beta)b + \beta c\theta.
 \end{aligned}$$

The wage is the Nash formula, not an arbitrary residual.

4.3 Verification and public declarations

Product-form job creation proves equation (1), while $V_R = 0$ proves (2). The computed surplus gives (3), worker values give (4), and the definition of U_R gives (7). Measure-form job destruction, expected excess, and search gain first derive candidate equation (8); continuation decomposition and the explicit wage then prove (5) and (6).

Principal declarations include:

- `Primitives.SatisfiesJobCreationProduct; ReducedEquilibrium.`
- `Primitives.positivePart_sub_integrable; Primitives.expectedExcess_eq_tailOptionValue.`
- `ValueEquilibrium.toReducedEquilibrium; ReducedEquilibrium.toValueEquilibrium.`
- `ReducedEquilibrium.toValue_toReduced; ValueEquilibrium.wage_eq_nash_formula.`
- `ValueEquilibrium.reconstruction_components; ValueEquilibrium.reconstruction_economic_roundtrip.`
- `valueEquilibrium_nonempty_iff_reducedEquilibrium_nonempty; m4_representation_capstone.`

4.4 Assumptions and exact scope

The forward bridge carries core, shock, and matching bundles; matching enters through M2 cutoff admissibility. Reverse reconstruction carries only core and shock bundles. The reverse constructor uses probability normalization and first-moment integrability, but not upper support. Upper support is needed for the paper-facing CDF-tail form of equation (10). Thus a full assumption bundle may occur in a signature while a proof term accesses only these named fields. Atomlessness, moment normalization, matching monotonicity, elasticity, and differentiability are unused.

The reduced round trip is exact. The value round trip proves equality of θ , V , U , J , W , wage, and surplus. Exact structure equality is not exported because proof-valued Bellman/admissibility fields are not the stable economic interface. Under the maintained bundles, nonemptiness is logically equivalent. Neither side is proved nonempty because no initial witness is constructed. M4 proves neither existence nor uniqueness.

Human-review conclusion. The robust reduced conditions and explicit Nash wage are economically correct. Equations (1)–(7) are verified rather than assumed; the two directions are non-circular; the reduced round trip is exact; and the value round trip preserves every economic object and surplus. The generic excess/tail analysis is valid, and no existence or uniqueness theorem is proved. M4 is **COMPLETE – GREEN**.

Dependency unlocked. M5 may study existence and uniqueness in the two-variable reduced system, knowing every admissible solution reconstructs the primitive value equilibrium.

5 M5: Existence and uniqueness of the static reduced equilibrium

Paper reference. Section 3, equations (10) and (13), Figure 1, and the uniqueness statement on journal page 402.

Adequacy status. Uniqueness: **COMPLETE – GREEN**. Existence: **COMPLETE – AMBER**. Overall M5: **COMPLETE – AMBER**.

5.1 Scalar curves and uniqueness

Define

$$\begin{aligned} H(d) &= \int \max\{x - d, 0\} dP, \\ N_{JD}(d) &= p + \sigma d - b + \frac{\lambda \sigma}{r + \lambda} H(d), \\ \theta_{JD}(d) &= \frac{N_{JD}(d)}{\beta c / (1 - \beta)}, \\ R(d) &= q(\theta_{JD}(d)) \frac{(1 - \beta) \sigma}{r + \lambda} (\varepsilon_u - d) - c. \end{aligned}$$

Positive part is one-Lipschitz, hence $|H(d_2) - H(d_1)| \leq |d_2 - d_1|$. Expected excess is therefore antitone and continuous. For $d_1 < d_2$, the increase in N_{JD} is bounded below by $\sigma r(d_2 - d_1)/(r + \lambda) > 0$, so the JD-implied tightness is strictly increasing. Strict decrease and positivity of q make the positive-tightness JC locus strictly downward sloping. These opposite slopes prove `ReducedEquilibrium.unique` without an existence assumption.

5.2 Explicit existence correction

Slopes alone do not imply an intersection. M5 adds exactly:

`StaticExistenceAssumptions.q_continuousOn_pos`: continuity of q on positive tightness; and
`lower_crossing`: a $d_0 < \varepsilon_u$ with $\theta_{JD}(d_0) > 0$ and $R(d_0) > 0$.

This bundle contains no root or equilibrium witness. Lean derives $R(\varepsilon_u) = -c < 0$, proves bracket continuity, and applies Mathlib’s `intermediate_value_Icc`’ to obtain a strictly interior root. The root constructs a reduced equilibrium. Thus existence is proved under explicit additional existence conditions, not under the original primitive assumptions alone.

The lower crossing is not circular: it assumes a positive residual rather than a root. It nevertheless supplies the missing endpoint condition. Figure 1 proves uniqueness from slopes but does not itself prove existence, so M5 is a corrected conditional-existence theorem.

5.3 Existence assumption and primitive upgrade path

The exact committed closure bundle is:

```
structure StaticExistenceAssumptions
(P : Primitives) : Prop where
q_continuousOn_pos :
ContinuousOn P.q (Set.Ioi 0)
lower_crossing :
∃ d0 : ℝ,
d0 < P.epsUpper ∧
0 < P.jobDestructionTheta d0 ∧
0 < P.staticCrossingResidual d0
```

`q_continuousOn_pos` is a regularity assumption: it makes the scalar residual continuous wherever JD-implied tightness is positive. `lower_crossing` is the substantive endpoint closure. It supplies the positive lower residual while Lean derives the upper residual as $-c < 0$. Because the lower sign is not currently derived from primitives, existence remains **COMPLETE – AMBER**.

A proposed, unformalized matching-boundary bundle would add $q(\theta) \rightarrow +\infty$ as $\theta \downarrow 0$, together with continuity of q on positive tightness. This Inada condition is not sufficient by itself. The separate primitive profitability condition $b < p + \sigma\varepsilon_u$ is needed to ensure that the JD curve reaches positive tightness. If `jobDestructionNet` is nonpositive everywhere, behavior of q on positive tightness cannot create an equilibrium.

The proposed five-step route is:

1. use the one-Lipschitz bound on expected excess to obtain a sufficiently low cutoff with negative JD net value;
2. use upper support and upper-job profitability to show positive JD net value at ε_u ;
3. use continuity and strict increase to obtain a unique zero-net, zero-tightness boundary cutoff below ε_u ;
4. use the right-hand Inada limit to obtain a positive scalar residual just above that boundary; and
5. construct `StaticExistenceAssumptions` and reuse `reducedEquilibrium_nonempty` A D X unchanged.

For $q(\theta) = \kappa\theta^{-\eta}$, with $\kappa > 0$ and $0 < \eta < 1$, vacancy filling is positive, strictly decreasing, and Inada at zero, while $\theta q(\theta)$ increases. A future M5b can certify these facts with `Real.rpow`, combine them with upper-job profitability, and upgrade existence to green for that specification. No such theorem is currently implemented.

The full economic argument, comparison table, proposed bundles, and future Lean theorem targets are in `docs/static_existence_foundation.md`.

5.4 Principal declarations and transport

- `Primitives.expectedExcess_antitone`, `expectedExcess_lipschitz`, and `expectedExcess_continuous`.
- `Primitives.jobDestructionNet`, `jobDestructionTheta`, `staticCrossingResidual`, and `staticCrossingResidual_strictAntiOn`.
- `jobDestruction_curve_strictMono` and `jobCreation_curve_strictAnti`.
- `ReducedEquilibrium.unique` and `reducedEquilibrium_atMostOne`.
- `StaticExistenceAssumptions`, `exists_staticCrossing`, `reducedEquilibrium_nonempty`, and `reducedEquilibrium_existsUnique`.
- `valueEquilibrium_nonempty`, `ValueEquilibrium.theta_eq`, `ValueEquilibrium.reservationCutoff_eq`, and `ValueEquilibrium.economically_unique`.
- `m5_static_equilibrium_capstone`.

Uniqueness accesses core sign restrictions; shock probability normalization and first-moment integrability; and vacancy-meeting positivity and strict decrease. Existence-only theorems take the core, shock, and static-existence bundles, but no matching bundle. They access only shock probability and first-moment integrability plus `q_continuousOn_pos` and `lower_crossing`. Combined uniqueness capstones retain matching.

M5 does not access upper support, atomlessness, shock mean or variance normalization, worker-meeting monotonicity, elasticity, or differentiability.

M4 reconstruction transports reduced nonemptiness to `ValueEquilibrium` and proves economic equality of theta, cutoff, values, wage, and surplus. Exact equality of proof-valued equilibrium records is not claimed.

M5 does not prove unconditional existence, equation (14), unemployment or vacancy stocks, comparative statics, or dynamics.

Human-review conclusion. The uniqueness and IVT proofs passed review. Uniqueness is **COMPLETE – GREEN**; conditional existence and overall M5 are **COMPLETE – AMBER**.

Remaining foundational gap. Derive `lower_crossing` from more primitive boundary/range conditions on q , or certify it for a standard Cobb–Douglas matching function. Until then, existence remains conditional. This does not block M6.

Dependency unlocked. M6 may add unemployment flow balance and equation (14) to the uniquely determined static pair.

6 M6: Unemployment flow balance and the full static steady state

Paper reference	Section 3; journal p. 403, equation (14); journal p. 404, flow discussion; Figure 2
Adequacy status	COMPLETE – AMBER
Human-review grade	Stock completion/equation (14): GREEN; uniqueness: GREEN; transported existence and overall M6: AMBER
Lean sources	<code>SteadyState/Flows.lean</code> , <code>Equilibrium/SteadyState.lean</code> , <code>SteadyState/Unemployment.lean</code> , and <code>SteadyState/FullEquilibrium.lean</code>

6.1 Strict destruction rule and paper CDF

The economic rule is strict, $\varepsilon < d$. Lean therefore defines

$$L_{<}(d) = \text{shock}((-\infty, d)), \quad s(d) = \lambda L_{<}(d), \quad f(\theta) = \theta q(\theta).$$

The flows are $D(d, u) = s(d)(1 - u)$ and $C(\theta, u) = f(\theta)u$. Under `ShockAssumptions.noAtoms`, `Primitives.strictShockBelow_eq_cdf` proves $L_{<}(d) = F(d)$. Atomlessness is used at this bridge between the strict destruction event and the paper's CDF notation; it was not needed for M3's layer-cake identity. Employment and vacancies are derived, not stored:

$$n = 1 - u, \quad v = \theta u.$$

6.2 Equation (14) and stock uniqueness

Flow balance is $s(d)(1 - u) = f(\theta)u$. Positive tightness and vacancy meeting imply $f(\theta) > 0$; nonnegative λ implies $s(d) \geq 0$. Thus $s(d) + f(\theta) > 0$, and the unique real solution is

$$u^*(\theta, d) = \frac{s(d)}{s(d) + f(\theta)}.$$

Lean proves $0 \leq u^* < 1$, direct flow balance, and stock uniqueness through the biconditional theorem

`Primitives.flow_balance_iff_unemployment_eq`.

Replacing the strict mass by the CDF yields paper equation (14):

$$u = \frac{\lambda F(d)}{\lambda F(d) + \theta q(\theta)}. \tag{14}$$

The paper-facing theorem is `SteadyStateEquilibrium.equation14`.

6.3 Full state, exact representation, and flows

`SteadyStateEquilibrium P` extends the reduced pair only with unemployment, its bounds, and flow balance. It stores neither employment, vacancies, equation (14), nor a value equilibrium. Every reduced state is completed by `ReducedEquilibrium.toSteadyStateEquilibrium`. Exact public round trips are

```

ReducedEquilibrium.toSteady_toReduced
SteadyStateEquilibrium.toReduced_toSteady
steadyStateEquivReduced.

```

The full state recovers primitive values through `SteadyStateEquilibrium.toValueEquilibrium`, preserving θ and $cutoff$. The accounting theorems establish

$$C = \theta q(\theta)u = q(\theta)v, \quad D = \lambda F(d)(1 - u), \quad C = D.$$

6.4 Zero separation and the literal ratio

The model permits $s(d) = 0$. Then equation (14) gives $u = 0$, and the derived identity gives $v = 0$; the literal ratio is undefined. Lean proves $v = \theta u$ definitionally for every state, but proves $v/u = \theta$ only under `HasPositiveSeparationAt d`, after deriving $0 < u < 1$ and $0 < v$. The declarations are `unemployment_pos_of_positiveSeparation`, `vacancies_div_unemployment_eq_theta`, and the zero-separation theorems.

6.5 Declarations, assumptions, and adequacy

- Strict mass/CDF: `Primitives.strictShockBelow` and `strictShockBelow_eq_cdf`.
- Hazards/flows: `jobSeparationRate`, `jobFindingRate`, `jobDestructionFlow`, `jobCreationFlow`, and `totalTransitionHazard_pos`.
- Stock formula: `steadyStateUnemployment`, `steadyStateUnemployment_flow_balance`, and `flow_balance_iff_unemployment_eq`.
- Full representation: `SteadyStateEquilibrium`, both exact round trips, and `steadyStateEquivReduced`.
- Uniqueness/existence: `SteadyStateEquilibrium.unique`, `steadyStateEquilibrium_nonempty`, `steadyStateEquilibrium_existsUnique`, and the separate `m6_stock_completion_capstone` and `m6_conditional_existence_capstone`.

Stock completion accesses only nonnegative λ , atomlessness for the CDF bridge, and positive vacancy meeting at positive θ . It does not use discounting, dispersion, bargaining, vacancy cost, moment integrability, upper support, normalization, matching antitonicity/worker-rate monotonicity, elasticity, or differentiability. Full-state uniqueness inherits reduced uniqueness. Full-state nonemptiness is transported from M5 and accesses `StaticExistenceAssumptions` only through its reduced witness.

Stock completion, equation (14), flow equality, exact representation, and full-state uniqueness introduce no new closure premise and are GREEN. M6 introduces no new existence closure assumption. Its AMBER existence grade is inherited entirely through

```

StaticExistenceAssumptions.lower_crossing
→ reducedEquilibrium_nonempty
→ toSteadyStateEquilibrium
→ steadyStateEquilibrium_nonempty.

```

The separate conditional-existence capstone makes this inherited premise visible; overall M6 is COMPLETE – AMBER.

Interiority is separate. Zero separation implies $u = v = 0$, so $v = \theta u$ remains valid but the literal ratio is unavailable. Positive separation is required only for the ratio theorem, not for stock completion or existence.

Human review judged equation (14), accounting, zero separation, exact representation, and uniqueness faithful and non-circular. The detailed record is `docs/static_steady_state_adequacy.md`. A future

primitive route would expose positive λ , lower-tail richness/support, and cutoff admissibility. The M5b route combines right-hand Inada behavior of q with upper-job profitability; proving it upgrades both M5 and inherited M6 existence without changing the M6 architecture.

M6 proves no unconditional existence, Beveridge-curve slope or convexity, comparative static, equation (15), dynamic, or M7+ theorem.

Dependency unlocked. M7 may study static comparative statics; M9 may reuse the flow residual for equation (15). M5b remains the route for upgrading existence.

7 M7: Static order comparative statics

Paper reference	Section 3, journal pp. 401–404; equations (10), (13), (14); Figures 1–2
Adequacy status	COMPLETE – GREEN
Lean sources	ComparativeStatics/ParameterChanges.lean, FixedTightness.lean, EquilibriumOrders.lean, FlowImplications.lean, and StaticComparativeStatics.lean

7.1 Parameter updates and order results

The immutable operations `withCommonProductivity`, `withUnemploymentIncome`, `withShockArrivalRate`, and `withDiscountRate` change exactly one primitive. They preserve the shock law and matching technology definitionally and provide explicit transports of core, shock, and matching assumptions. No static-existence transport is defined.

At fixed tightness, higher p strictly lowers the cutoff, higher b strictly raises it, higher λ weakly lowers it, and higher r weakly raises it. Strict refinements recover strictness under positive expected excess; the strict r refinement also requires $\lambda > 0$. For supplied reduced equilibria, higher p lowers the cutoff and raises tightness; higher b raises the cutoff and lowers tightness; higher λ lowers the cutoff; and higher r lowers tightness. The exact declarations are

- `cutoff_strictAnti_commonProductivity_at_fixed_theta`, `cutoff_strictMono_unemploymentIncome_at_fixed_theta`;
- `cutoff_antitone_shockArrival_at_fixed_theta`, `cutoff_monotone_discountRate_at_fixed_theta`;
- `cutoff_strictAnti_shockArrival_at_fixed_theta_of_expectedExcess_pos`, `cutoff_strictMono_discountRate_at_fixed_theta_of_optionValue_pos`;
- `ReducedEquilibrium.order_of_commonProductivity`, `order_of_unemploymentIncome`;
- `ReducedEquilibrium.cutoff_strictAnti_shockArrival`, and `theta_strictAnti_discountRate`.

7.2 Hazards, flows, and stocks

Strict lower-tail inclusion gives weak separation ordering without atomlessness; matching monotonicity gives weak finding ordering. At a common current $0 \leq u \leq 1$, the hazard orders imply the corresponding initial job-creation and job-destruction flow orders. The algebraic theorem `steadyUnemployment_monotone` then proves that lower separation and higher finding weakly lower $s/(s + f)$.

The full-state declarations `aggregateProductivity_capstone` and `unemploymentIncome_capstone` combine strict cutoff/tightness results with weak separation, finding, unemployment, and employment results. Their companion `aggregateProductivity_initialFlows` and `unemploymentIncome_initialFlows` theorems hold unemployment fixed. The combined end-points are `m7_aggregateNetProductivity_capstone` and `m7_otherParameters_capstone`.

Unemployment effects are weak because the CDF may be flat and the worker meeting rate is only weakly monotone. Positive shock mass between cutoffs or strict worker-rate monotonicity would support strict refinements. Vacancies equal $v = \theta u$, whose factors move oppositely, so no vacancy sign is claimed.

7.3 Assumptions, adequacy, and deferrals

The public equilibrium signatures carry core, shock, and matching bundles. Proof terms use positive JD/JC coefficients, probability and first-moment facts through expected-excess monotonicity, positive and strictly decreasing q , and—for flows—weak worker-meeting monotonicity. The generic strict- tail lemma needs only finite measure, not atomlessness. Upper support and no atoms may be carried by an existing bundled JD theorem without being accessed by the M7 step. Shock normalization, matching elasticity, differentiability, and `StaticExistenceAssumptions` are unused.

Every core result is universally conditional on supplied equilibrium witnesses. M7 proves no existence theorem and therefore does not inherit M5’s AMBER grade. A future existence-backed selector would be separately AMBER, inherited from M5.

Human review found the strict and robust weak result decomposition mathematically and economically sound. M7 is COMPLETE – GREEN. It proves universal conditional results for supplied equilibria, not their existence; that does not inherit M5’s AMBER grade. A selected-equilibrium wrapper using `StaticExistenceAssumptions` would be separately AMBER.

Human-review checklist: ☑ Human mathematical/economic review is complete.

The equilibrium tightness response to λ , equilibrium cutoff response to r , all σ results, equation (11), and Appendix equations (A1)–(A12) remain M8. The durable matrix is `docs/static_comparative_statics_scope.md`.

8 M8: Appendix derivative comparative statics

Paper reference	Appendix, equations (A1)–(A12), and equation (11)
Implementation status	COMPLETE – AMBER ; reviewed 2026-08-01
Adequacy status	analytic infrastructure: COMPLETE – GREEN ; supplied-path results: COMPLETE – AMBER ; overall: COMPLETE – AMBER
Lean sources	MP1994V2/Assumptions/Appendix.lean and the nine M8 files under MP1994V2/ComparativeStatics/

8.1 Exact theorem contract

M8 proves the Appendix identities and signs for supplied objects satisfying `LocalReducedEquilibriumPath` or `FixedTightnessSigmaPath`. These structures store differentiability and exact local JD/JC closure equations, but no derivative signs or Appendix conclusions. Thus the results are non-circular. M8 does not construct a locally differentiable equilibrium selection by an implicit-function theorem; this is the sole AMBER gap.

The public interface comprises equation (11), equations (A1)–(A12), the lambda cutoff and tightness signs, the negative discount-rate tightness sign and exact (A8) cutoff-sign equivalence, and the positive dispersion signs under $b \leq p$. The capstones are `FixedTightnessSigmaPath.m8_fixedTightness_capstone`, `LambdaEquilibriumPath.m8_lambda_capstone`, `DiscountEquilibriumPath.m8_discount_capstone`, `DispersionEquilibriumPath.m8_dispersion_capstone`, and `m8_appendix_capstone`.

The individual public declarations are `LambdaEquilibriumPath.equationA1` through `equationA4`, `cutoffSlope_neg`, and `thetaSlope_neg`; `DiscountEquilibriumPath.equationA5` through `equationA8`,

thetaSlope_neg, and cutoffSlope_pos_iff_A8_rhs_pos; and DispersionEquilibriumPath.equationA9 through equationA12, thetaSlope_pos, cutoffSlope_pos_iff_A12_expression_pos, and cutoffSlope_pos_of_b_le_p. Cross-multiplied, normalized, coefficient-sign, and slope-equation forms remain public as named in the Lean sources and audit.

8.2 Analytic and algebraic foundation

AppendixMatchingAssumptions gives differentiability of q on positive tightness and matching elasticity in $(0,1)$. cdf_continuous derives CDF continuity from atomlessness, and hasDerivAt_expectedExcess proves the required FTC derivative. The strict upper-tail fact for (A4) uses atomlessness and the almost-sure upper bound. The moment identity for (A11) visibly takes ShockNormalizationAssumptions. Scalar transformations are isolated in AppendixAlgebra.lean.

Low-level lambda results use the core boundary convention $0 \leq \lambda$; the capstone explicitly takes $0 < \lambda$, matching the paper's interior convention. The discount cutoff response is intentionally ambiguous and is characterized by the exact (A8) condition. Dispersion raises both derivatives under the explicit condition $b \leq p$.

8.3 Assumptions and fields accessed

CoreEconomicAssumptions supplies positive/nonnegative primitive signs and the resulting nonzero denominators. ShockAssumptions supplies the probability law, atomlessness, almost-sure upper bound, and first-moment integrability. ShockNormalizationAssumptions supplies zero mean and unit second moment for the (A11) tightness result, but is not used by cutoffSlope_pos_of_b_le_p. MatchingAssumptions supplies positive, hence nonzero, q at positive tightness. AppendixMatchingAssumptions supplies q_differentiableOn_pos, elasticity_pos, and elasticity_lt_one for $\eta(\theta) = -\theta q'(\theta)/q(\theta)$. Path records supply derivative witnesses, positivity/admissibility, and eventual exact JD/JC closure; they supply no sign, Jacobian condition, or existence witness.

8.4 Dependency and adequacy audit

No M8 theorem imports or takes StaticExistenceAssumptions, and no M5 witness is selected. The AMBER grade therefore reflects only the supplied-path premise, not M5's global existence qualification. No M8b implicit-function, Section 4, cyclical, or Markov result is included. A worker-meeting-rate derivative is optional and deferred. The precise boundary and six-step M8b route are in docs/appendix_differentiability_scope.md.

Human mathematical/economic review confirmed that the Appendix identities and sign conclusions are faithful, that the path premise is non-circular, and that the AMBER source is supplied differentiable local paths rather than M5. The paper-facing lambda capstone correctly states $0 < \lambda$; M8b is the route from path-conditional AMBER to GREEN.

Human-review checklist: ☒ Human mathematical/economic review is complete (2026-08-01).

9 M8b.1: Local IFT foundation and lambda path

Status: COMPLETE – GREEN, reviewed 2026-08-01.

M8b.1 proves local C^1 regularity of the fixed-tightness sigma residual and lambda scalar-crossing residual at a supplied $R : \text{ReducedEquilibrium } P$. Their base roots follow from the stored JD and JC equations. The cutoff partials are derived rather than assumed: the fixed-sigma partial is strictly positive and the lambda crossing partial is strictly negative.

The generic exists_localImplicitCutoffPath wraps Mathlib's ContDiffAt.implicitFunction. It builds scalar invertibility from a proved nonzero derivative and exposes local C^1 regularity, residual closure, and graph uniqueness. The economic constructors are ReducedEquilibrium.exists_fixedTightnessSigmaPath_of_ift

and `ReducedEquilibrium.exists_lambdaEquilibriumPath_of_ift`; their selected paths feed the reviewed M8 capstones.

Human review graded the wrapper, both constructed paths, and the combined M8b.1 result COMPLETE – GREEN relative to a supplied reduced equilibrium. Base roots and nondegeneracy are proved, and no derivative sign is assumed. No theorem uses `StaticExistenceAssumptions`, selects an M5 equilibrium, or assumes a root, nondegeneracy, or derivative sign. `AppendixIFTAssumptions` contains only C^1 regularity of primitive q on positive tightness. This is an explicit technical strengthening, not a paper conclusion. No global path or equilibrium uniqueness is claimed. At M8b.1 completion, overall M8 remained COMPLETE – AMBER because discount and dispersion paths were still supplied. M8b.2 now constructs them and has passed review.

Human-review checklist: ☒ Human mathematical/economic review is complete (2026-08-01).

10 M8b.2: Discount and dispersion IFT paths

Status: COMPLETE – GREEN, reviewed 2026-08-02.

For each supplied $R : \text{ReducedEquilibrium } P$, M8b.2 parameterizes the scalar JD/JC crossing residual by the discount rate and by dispersion. The base roots and JD-implied base tightness are derived from the equations in R . Both cutoff partials are proved strictly negative by reduction to `Primitives.deriv_staticCrossingResidual_neg`; no Jacobian or nondegeneracy condition is assumed.

The local C^1 residual theorems feed `exists_discountEquilibriumPath_of_ift` and `exists_dispersionEquilibriumPath_of_ift`. These constructors prove local parameter positivity, JD, product-form JC, base-point recovery, and differentiability before creating the existing M8 path records. Their selected paths feed `m8b_discount_capstone` and `m8b_dispersion_capstone`. The discount result proves falling tightness and the exact A8 iff cutoff-sign characterization. Normalization and $b \leq p$ enter only the dispersion sign capstone, not path construction.

`ReducedEquilibrium.m8b_full_appendix_capstone` combines all four IFT-derived Appendix paths. It takes no path object, `StaticExistenceAssumptions`, M5 selection, derivative sign, or nondegeneracy assumption. Human review found the discount residual/path, dispersion residual/path, and full capstone COMPLETE – GREEN. The existing M8 equations and signs are reused unchanged.

The full Appendix IFT-derived package is COMPLETE – GREEN relative to a supplied regular reduced equilibrium, Appendix matching and IFT regularity, and the explicitly stated sign conditions. The historical AMBER grade for the original supplied-path M8 layer remains accurate. Selecting the base equilibrium from primitives would be a separate wrapper and would inherit M5's AMBER existence qualification.

Human-review checklist: ☒ Human mathematical/economic review is complete (2026-08-02).

11 M9.1: Anticipated two-state foundations

Status: COMPLETE – GREEN, reviewed 2026-08-02.

M9.1 defines an involutive recession/boom process. Boom changes only common productivity from p to p_H ; all remaining primitives and the idiosyncratic shock law are shared. `TwoStateValueEquilibrium P T` records state-contingent primitive value equations with the aggregate switching rate included explicitly.

Surplus is derived as $S_s(\varepsilon) = J_s(\varepsilon) + W_s(\varepsilon) - U_s$. The three continuation expressions are defined once and reused. From a supplied equilibrium Lean derives firm sharing, zero vacancy values, the statewise vacancy-cost identity, and the raw surplus flow equation. Under $D : \text{ShockAssumptions } P$, the general coupled system is

$$(r + \lambda + \mu)S_s(\varepsilon) = p_s + \sigma\varepsilon - b + \lambda \int \max\{S_s(x), 0\} dF(x) - \beta\theta_s q(\theta_s)S_s(\varepsilon_u) + \mu \max\{S_{s'}(\varepsilon), 0\}.$$

Here integration against dF is represented by integration against `P.shock`. Probability normalization is installed locally from `ShockAssumptions`; it is not stored in the equilibrium structure.

The state-specific theorems retain the other state's positive part. Equations (16)–(18) are proved only under explicit pointwise sign hypotheses. The result is conditional on a supplied `TwoStateValueEquilibrium`; it does not prove existence. No cutoff, cutoff ordering, affine result, reduced two-state equilibrium, job-creation system, unemployment dynamics, or impact asymmetry is introduced. See `docs/two_state_foundations_scope.md`.

The paper writes equations (16)–(18) after using the expected ordering of boom and recession cutoffs. M9.1 instead formalizes one general max-based system. M9.2 must prove the ordering before interval-specific formulas are certified.

Human review confirmed that aggregate capital gains appear in all four primitive value equations and that the coupled surplus equation is derived, not assumed. Equations (16)–(18) are max-form sign reductions, not yet the cutoff-indexed interval-integral formulas. Conditionality on a supplied equilibrium is the correct scope and does not make M9.1 AMBER. No cutoff or ordering assumption is present.

Human-review checklist: ☒ Human mathematical/economic review is complete (2026-08-02).

12 M9.2: Statewise cutoffs and ordering gate

Status: **COMPLETE – GREEN**, conditional on a supplied `TwoStateValueEquilibrium` (human review 2026-08-04).

M9.2A is **COMPLETE – GREEN**, reviewed 2026-08-02; M9.2B is **COMPLETE – GREEN**, reviewed 2026-08-04; M9.2C is **COMPLETE – GREEN**, reviewed 2026-08-04.

The ordering-independent layer derives the two coupled surplus-increment identities, simultaneous strict monotonicity, quantitative lower and upper increment bounds, global Lipschitz continuity, positive upper-support surplus, and a negative-surplus witness in each state. It constructs the unique zero of each actual statewise surplus as `TwoStateValueEquilibrium.reservationCutoff` and proves the full surplus and firm-value reservation property.

The exact measure-form continuation identity is

$$\int S_s^+(x) dF(x) = \int_{[d_s, \varepsilon_u]} S_s(x) dF(x).$$

Here `P.shock` is the probability law, and integration with respect to the induced CDF F is represented in Lean by integration against `P.shock`. The almost-sure upper bound and zero surplus at the cutoff justify the closed interval. No cross-state order is used.

The initial adequacy attempt did not derive $d_B < d_R$. Under the reverse geometry, free entry implies an endogenous tightness order, but subtraction of the two state equations retains a search-gain difference whose magnitude is not controlled by the current matching assumptions. That term competes with the positive productivity gap and signed continuation differences, so no contradiction follows. A separately justified bound on the cross-state outside-option/search-gain response is the weakest plausible additional condition identified for review; it was not added to Lean.

The implemented maximum-principle route assumes the reverse order only inside a contradiction proof. It proves boom surplus is globally below recession surplus and uses free entry to order tightness. Regional gap constancy gives $D_u \leq S_B^+(x) - S_R^+(x)$ and the corresponding integrated bound. The separately exposed upper-support equation is

$$(r + \lambda + 2\mu)D_u = (p_H - p) + \lambda(I_B - I_R) + \frac{\beta c}{1 - \beta}(\theta_R - \theta_B).$$

It implies $(r + 2\mu)D_u \geq p_H - p > 0$, contrary to $D_u \leq 0$. Lean therefore proves `boom_cutoff_lt_recession_cutoff` without adding an order, surplus, tightness, search-gain, or closure assumption, and derives strict boom upper-surplus and tightness orders.

M9.2B is **COMPLETE – GREEN**, reviewed 2026-08-04.

M9.2C derives exact ordered equations (16)–(18) only after applying the proved order $d_B < d_R$. Coupled two-point identities yield the common upper-region slope $\sigma/(r + \lambda)$ and the lower boom slope $\sigma/(r + \lambda + \mu)$

between cutoffs. The pointwise consequences include equation (23). Global recession and boom hinge identities integrate to expected-excess and exact split CDF-tail option values. Substitution and statewise free entry then prove equations (19), (20), (22), and (24), with the aggregate transition gap entering equation (24) negatively.

The public capstones are `m9_2C_ordered_equations_capstone` and `m9_2_full_cutoff_capstone`. They are conditional on a supplied `TwoStateValueEquilibrium`; no cutoff order, affinity, derivative, or numbered equation is assumed or stored in the equilibrium. No `StaticExistenceAssumptions` or primitive-selected two-state equilibrium is used. Human review on 2026-08-04 graded M9.2C and full M9.2 **COMPLETE – GREEN**, conditional on the supplied equilibrium. It confirmed the exact ordered equations, stronger difference-form equation (21), both hinge and option-value identities, equations (19), (20), (22), (23), and (24), and both capstones. No M9.3 unemployment, job-creation, or impact-asymmetry theorem was begun. See `docs/two_state_cutoff_scope.md` and `docs/two_state_cutoff_ordering_analysis.md`, plus `docs/two_state_ordered_equations_scope.md`.

M9.2A human-review checklist: ☑ Human mathematical/economic review is complete (2026-08-02).

M9.2B human-review checklist: ☑ Human mathematical/economic review is complete (2026-08-04). Review confirmed $d_B < d_R$, the global reverse-order surplus comparison, free-entry tightness comparison, integrated active-gap bound, and upper-gap contradiction. No order, affinity, derivative, search-gain, or closure assumption is present.

M9.2C/full-M9.2 human-review checklist: ☑ Human mathematical/economic review is complete (2026-08-04).

13 M9.3: Job creation, unemployment dynamics, and impact asymmetry

Status: **COMPLETE – GREEN**, reviewed **2026-08-04**, conditional on a supplied `TwoStateValueEquilibrium` and admissible employment distributions where impact accounting is stated.

Job creation and equations (25)–(30)

`equation25` and `equation26` state the two upper-region surplus equations after the reviewed order $d_B < d_R$ has been derived. `equation27` exposes the reviewed recession affine formula as the paper’s solution. `equation28` derives recession free entry, while `equation29` gives boom surplus as its own discounted productivity gap plus anticipated recession value. The definition `boomJobCreationDenominator` is proved by `boomJobCreationDenominator_eq` to equal $(r + \lambda)S_B(\varepsilon_u)$, hence is positive by `boomJobCreationDenominator_pos`. `equation30` gives boom free entry. `boomJobCreationDenominator_lt_staticGap` and `staticJobCreationTarget_lt_boomVacancyMeetingRate` isolate the anticipation wedge. `m9_3_jobCreation_capstone` collects these results.

Hazards, flows, and equation (15)

The vacancy contact rate $q(\theta)$, worker meeting hazard $\theta q(\theta)$, and aggregate creation flow $u\theta q(\theta)$ are distinct. `idiosyncraticDestructionHazard`, `workerMeetingHazard`, `jobCreationFlow`, and `idiosyncraticDestructionFlow` encode the statewise rates. Under core matching and Appendix matching assumptions, `workerMeetingHazard_strictMonoOn_pos` proves strict monotonicity on positive tightness. No IFT assumption is used.

The vector field `unemploymentDrift` satisfies

$$G_s(u) = (1 - u)\lambda F(d_s) - u\theta_s q(\theta_s) \quad (15)$$

by `equation15`. This is a model law, not a consequence of the Bellman system. `SatisfiesEquation15` describes a supplied differentiable path; M9.3 proves no ODE existence theorem. `stationaryUnemployment`, its bounds, zero-drift result, and drift-sign theorems characterize the fixed-state affine field.

`boom_unemploymentDrift_lt_recession` compares the two fields at a common interior unemployment stock. The capstone is `m9_3_unemployment_capstone`.

Aggregate-shock impact accounting

`StateEmploymentDistribution` stores a finite employment measure of mass at most one supported above the current cutoff. It assumes neither a density nor positive mass between cutoffs. `afterAggregateShock` transparently restricts this measure above the new cutoff. It contains no creation component; `impactCreationMass_eq_zero` records that timing convention.

The upturn theorem `recession_to_boom_employmentMeasure_eq` leaves a recession-supported measure unchanged. The downturn theorem `boom_to_recession_measure_decomposition` splits the old measure into survivors and the interval $[d_B, d_R)$, whose exact destruction mass is given by `boom_to_recession_impactDestruction_eq`. The strict result `boom_to_recession_employmentMass_lt` additionally assumes positive pre-shock mass in this interval. Real unemployment changes are completed by `boom_to_recession_unemploymentMass_eq` and `recession_to_boom_unemploymentMass_eq`.

`m9_3_impact_asymmetry_capstone` and `m9_3_full_cyclical_dynamics_capstone` collect the impact and full interfaces. No cutoff order, affinity, asymmetry, derivative sign, `StaticExistenceAssumptions`, or primitive-selected equilibrium is assumed. No statistical volatility, lead-lag, simulation, equation (31), or M10 theorem is claimed. See `docs/two_state_cyclical_dynamics_scope.md`.

Human mathematical/economic review confirmed that equations (25)–(30) agree with Section 4 and that equation (27) appropriately reuses the stronger M9.2C affine theorem. Equation (15) is a transparent labor-flow vector field; no ODE existence or global path theorem is claimed. Vacancy contact, worker meeting, and aggregate creation rates remain distinct. The impact operator is an incumbent-first timing convention: downturn destruction is exactly $[d_B, d_R)$, strictly only under positive interval mass, while an upturn has no instantaneous creation. No statistical volatility, lead-lag, or simulation claim is present.

The full M9.1–M9.3 Section 4 productivity-shock package is **COMPLETE – GREEN**, conditional on a supplied `TwoStateValueEquilibrium`. Equation (31) remains unimplemented.

Human-review checklist: ☒ Human mathematical/economic review is complete (2026-08-04).

14 M10.1: Finite-state Markov equilibrium, equations (32)–(34)

Status: **COMPLETE – GREEN**, reviewed 2026-08-04.

`FiniteAggregateProcess` stores state productivity, a nonnegative continuous-time aggregate-arrival rate, and finite transition weights. `FiniteAggregateProcessAssumptions` requires nonnegative weights and unit row sums. `nextExpectation` is their finite weighted sum, with constant, additive, scalar, nonnegative, and monotone laws proved independently of equilibrium.

`FiniteMarkovCandidate` stores only tightness, cutoff, and raw surplus; worker meeting, active surplus, and the surviving-surplus integral are derived. `FiniteMarkovEquilibrium` records admissibility and equations (32)–(34) for a supplied equilibrium. `equation32`, `equation33`, and `equation34` expose the equations. `upperSurplus_pos`, `workerMeetingHazard_pos`, and `aggregateContinuation_nonneg` give immediate sign consequences. The generic capstones are `m10_1_finiteMarkov_capstone` and `m10_1_finiteState_foundations_capstone`.

`TwoStatePrimitives.toFiniteAggregateProcess` has a deterministic-other-state kernel, whose expectation reduces to the other state. Every supplied M9 `TwoStateValueEquilibrium` embeds through `toFiniteMarkovEquilibrium` with identical tightness, derived cutoff, and actual surplus. The two embedding capstones record this forward specialization.

This is a representation theorem conditional on supplied equilibrium objects; it proves no existence or uniqueness. `P.lambda` and `aggregateArrival` are continuous-time rates. Equation (35), employment transitions, equations (36)–(38), density assumptions, and numerical work are absent. See `docs/finite_markov_equilibrium_scope.md`.

Human review confirmed that aggregate arrival is a continuous-time marked-Poisson rate; transition weights form conditional next-mark distributions with nonnegative unit-sum rows; equation (32) distinguishes vacancy and worker contact rates; and equation (34) contains the exact finite active-surplus expectation. The deterministic-other-state M9 embedding is faithful, and no existence, uniqueness, cutoff-sign, or rate-as-probability shortcut is present.

Human-review checklist: ☒ Human mathematical/economic review is complete (2026-08-04).